

HOSPITAL APPOINTMENT AND BILLING ANALYTICS

Given Scenario:

A hospital wants to analyze:

- Patients
- Doctors
- Departments
- Appointments
- Treatments
- Bills
- Payments

The goal is to build a MySQL database and generate reports for hospital management.

Database Design:

The Hospital Appointment and Billing Analytics system is designed to manage patient appointments, treatments, billing, and payments. The database supports hospital operations and provides analytical reports for monitoring and management. This database contains seven tables:

Tables	Description
patients	Stores patient information
doctors	Stores doctor details
departments	Stores department information
appointments	Stores appointment records
treatments	Stores treatment details
Bills	Stores billing information
payments	Stores payment transactions

Table Relationships:

Departments → Doctors: The departments table is connected to the doctors table through department_id.

Patients → Appointments: The patients table is connected to the appointments table through patient_id.

Doctors → Appointments: The doctors table is connected to the appointments table through doctor_id.

Appointments → Treatments: The treatments table is connected to the appointments table through appointment_id.

Appointments → Bills: The bills table is connected to the appointments table through appointment_id.

Bills → Payments: The payments table is connected to the bills table through bill_id.

Key Insights from Reports:

- The Cardiology department generates significant revenue through consultations and treatments.
- Hyderabad has the highest number of patients in the database.
- UPI is one of the most commonly used payment methods.
- A few bills have pending or partial payments, helping identify outstanding dues.
- Appointment status reports help to track cancelled and pending appointments, supporting operational planning.
- Certain doctors handle a higher number of appointments, indicating greater patient demand.
- Department-wise revenue reports help identify the hospital's top-performing departments.